Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building public-facing **Python** APIs and web applications with **Django**, Flask, and FastAPI on **PostgreSQL**, with automated testing, code reviews, release pipelines, and Linux-based deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant **PostgreSQL** schemas with indexing and query tuning for 200+ users and 10K+ records, cutting latency 35%, and enforced row-level isolation across a 500+ user platform.
- Built **Python** FastAPI services and background workers with explicit job states and failure isolation, processing 1K+ jobs at 3x throughput and shipping through **CI/CD** releases at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a modular **Flask** REST backend organized into Blueprints for auth, dashboards, and grading, with Flask-Login sessions and MySQL persistence, serving 150+ instructors and 10K+ daily requests.
- Fixed production bugs such as OCR timeouts with fallback routing and multithreaded batch processing for 1K+ files per run, keeping APIs stable at peak load and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Maintained **release pipelines** in Jenkins with feature, staging, and production branches, building and installing executables on Linux, Windows, and macOS VMs nightly with 20+ **automated test** suites.
- Modernized legacy API integrations with object-oriented refactoring, pagination, field filtering, and caching, improving response times 30% and cutting technical debt 20% with a distributed Microsoft and GitHub team.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **Django** resource-management platform with REST APIs, a PostgreSQL data model, and real-time WebSocket updates for allocations, billing, and leads, syncing 2M+ records across multiple locations in real time.
- Monitored health with Sentry, Azure Monitor, and Kibana, and took part in **code review** and rollout across 6+ workflows with PyTest and Selenium tests, reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built public-facing **Flask APIs** for bookings, pricing, coupons, and payments with JWT authentication, keeping core responses under 200ms across 50K+ records for Android, iOS, and web clients.
- Automated refunds and disputes through idempotent payment webhooks, cutting manual handling about 70%, and presented the end-to-end architecture to the client before implementation began, after mapping 12+ integration points.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, Bash, Shell
Backend: Django, Flask, FastAPI, REST APIs, Microservices, Background Jobs, JWT, Webhooks
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, Schema Design, Query Optimization, Indexing
Platform & Ops: Linux, Docker, Kubernetes, AWS, Azure, Sentry, Kibana, Azure Monitor
Quality: PyTest, Selenium, Unit Testing, Integration Testing, Code Reviews, CI/CD, Jenkins, Git, Agile, GitHub Actions

Projects

Linux Fleet Inventory Service | *Python, Django REST, PostgreSQL, Celery, Ubuntu*

- Designed a **Django** service where lightweight agents on **Ubuntu** machines report installed packages, pending security updates, and hardware inventory, with an API for querying and scheduling upgrades.

Package Compliance Reporter | *Python, Flask, PostgreSQL, SQLAlchemy, pytest*

- Designed a **Flask** API that compares each machine's packages against security advisories, flags vulnerable versions, and produces per-team **compliance reports** backed by SQLAlchemy models and automated pytest suites.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, MySQL, Docker, Linux*

- Extended an **open-source** database workspace supporting 10+ engines including PostgreSQL across Windows, macOS, and Linux, with schema synchronization and AI-generated SQL reaching 82% accuracy, set up with Docker and Maven.